

Viva Q&A — Most Likely Questions

These are the ones examiners almost always ask. Know these and you'll be fine.

1. Why did you choose this topic?

“Phishing is the number one cybercrime globally — over 4.7 million attacks in 2023. I wanted to do something practical in cybersecurity. The existing defences are either too slow or always behind. I found a gap in the literature — no one had done a controlled comparison of classifiers using only structural URL features on a large dataset with explicit FPR reporting.”

2. What is your research question?

“Can machine learning models trained on lexical URL features alone achieve high accuracy and low False Positive Rates for real-time phishing detection? The answer is yes — but only with non-linear ensemble methods like Random Forest.”

3. How did you collect and validate your data?

“I used a publicly available Kaggle dataset — 641,053 URLs after cleaning. I removed duplicates, handled missing values, and used stratified $80/20$ splitting to maintain class balance. Fixed random seed of 42 for full reproducibility. Cross-validation confirmed stability with only 0.17% standard deviation.”

4. Why these three classifiers?

“One from each family. Logistic Regression — linear baseline. Random Forest — non-linear ensemble. SVM — second linear model to confirm whether the limitation is the algorithm or the linearity. The fact that LR and SVM got almost identical results — 78.94% and 78.43% — proves the bottleneck is linearity itself.”

5. Explain your methodology.

“CRISP-DM framework mapped to Agile sprints. Positivist philosophy, deductive approach. I engineered 10 lexical features from the URL string, trained three classifiers on the same data and features, and evaluated using six metrics. It is a controlled experiment — same data, same features, same split — only the algorithm changes.”

6. What are your main findings?

“Random Forest: 94.03% accuracy, F1-Score 0.9120, FPR 5.45%. Linear models: around 78% accuracy, 50% recall — they miss half the malicious URLs. The 15-percentage-point gap proves the relationship between URL features and phishing is non-linear. Cross-validation confirms stability.”

7. Why did Random Forest outperform the linear models?

“Phishing patterns are non-linear. A URL with many dots is not necessarily malicious — `bbc.co.uk` has three dots. But many dots combined with a long path and high entropy is very suspicious. Linear models cannot capture these feature interactions. Random Forest builds 200 trees that each split on different features, so it can model complex combinations.”

8. What does 5.45% FPR mean in practice?

“Out of every 100 legitimate URLs, about 5 get incorrectly flagged. That is roughly 1 in 18. For a first-layer filter combined with other security measures, this is acceptable. It meets my non-functional requirement of FPR below 10%.”

9. What are the limitations of your study?

“Four. Temporal drift — the model may degrade as phishing tactics evolve. Lexical features alone cannot catch perfect mimicry. I only tested three algorithms — not gradient boosting or deep learning. And the 5.45% FPR, while acceptable, is not negligible in high-traffic environments.”

10. What would you do differently?

“Temporal validation — train on older URLs, test on newer ones. Add character n-grams to catch brand spoofing. Test XGBoost or LightGBM. And deploy as a browser extension to measure real-world latency and user experience.”

11. How does your work contribute to existing knowledge?

“Three contributions. Largest controlled lexical comparison on 641,000 URLs. Explicit FPR reporting which most studies omit. And a fully reproducible open-source pipeline on GitHub.”

12. What challenges did you face?

“Malformed URLs crashing the parser — solved with vectorised pandas operations. Class imbalance — solved with balanced class weights. Computational cost of cross-validation — solved with balanced subsampling of 100,000 rows.”

13. Explain how PhishGuard works.

“React web app. User submits a URL, JavaScript extracts the same 10 features, passes them to a scoring function that replicates the Random Forest logic. Displays a risk score 0–100 with colour-coded verdict and feature breakdown. Runs entirely in the browser — no server, no API calls, instant results.”

14. What ethical considerations did you address?

“Low risk — no human participants, no personal data. All URLs treated as text strings in an offline environment. Publicly available dataset. Ethics form HE35 submitted December 2025 and approved. Followed NCG Research Ethics Policy.”

15. What did you learn from this project?

“That empirical testing matters more than assumptions — I expected linear models to be decent but they were poor. That Agile sprints prevent scope creep. And that documenting negative results — why linear models failed — was one of the most informative parts of the work.”

Numbers to Glance At Before You Walk In

Metric	Value
Dataset	641,053 URLs
RF Accuracy	94.03%
RF F1-Score	0.9120
RF FPR	5.45%
LR Accuracy	78.94%
SVM Accuracy	78.43%
Top feature	dot_count (F = 31,994)
Cross-val	91.95% $\pm$ 0.17%
Trees	200, depth 20